

PHSDSC717P – Mathematical Methods IV Lab

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Part I

Assignments

1 Complex Roots

- (a) Using the two-dimensional Newton-Raphson method, find the cube roots of a complex number by forming two simultaneous equations from the real and imaginary parts.
- (b) Calculate the roots by de Moivre's formula and compare the results.
- (c) Find complex roots of an arbitrary algebraic equation.

HINTS AND RELEVANT PYTHON FUNCTIONS:

- For obtaining the modulus and argument of a complex number, you may use `cmath.polar`.
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2 Improper Integrals

Consider the integrals, $I_c(R) = \oint_C f(z)dz$ and $I_r(R) = \int_{-R}^R f(x)dx$, C being a closed contour going from $-R$ to R along the real axis and coming back along a semi-circular contour of radius R centered at the origin and on the upper half of the complex plane. For each of the functions, (i) $f(z) = \frac{1}{z^2 + 1}$ (ii) $f(z) = \frac{e^{3iz}}{z^4 + 81}$ (iii) $f(z) = \frac{e^{5iz}}{z^6 + 64}$,

(a) evaluate $\int_{-\infty}^{\infty} f(x)dx \equiv \lim_{R \rightarrow \infty} I_r(R)$,

(b) calculate the contribution from the semi-circular arc, $I_{sc}(R)$, as $R \rightarrow \infty$, and,

(c) calculate $\lim_{R \rightarrow \infty} I_c(R)$.

HINTS AND RELEVANT PYTHON FUNCTIONS:

- Use Simpson's $\frac{1}{3}$ rule for the integrals by writing your own code.
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3 Gram-Schmidt Orthogonalization

- (a) Write a function which takes a pair of n -component vectors as input and returns a pair of orthogonal vectors using Gram-Schmidt orthogonalization. Write a function to check the orthogonality of the vectors. (Your function should work for vectors with real or complex elements.)
 - (b) Write a function which takes any number of n -component vectors as input and returns the same number of orthogonal vectors using Gram-Schmidt orthogonalization. Write a function to check the orthogonality of the vectors by forming a suitable orthogonal/unitary matrix. Test your functions by reading vectors from rows of an input file and writing the output to another file.
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4 Solution of Linear Systems of Equations using Gauss elimination and Gauss-Jordan methods

- (a) Write a function to calculate the determinant of a matrix using the Gauss elimination method.
- (b) Write a function to find the solution of linear system of equations, $Ax = b$, using the Gauss elimination method. The function should return an error message if the determinant of A vanishes.
- (c) Redo the above task using the Gauss-Jordan method.
- (d) Write a function to calculate the inverse of a matrix using the Gauss-Jordan method.

Note: Your codes should work for linear systems of arbitrary size.

Part II

Theory

Reference: Press et. al., *Numerical Recipes in Fortran*

1 Two-dimensional Newton-Raphson Method

Problem: Find a root of $\vec{F}(\vec{x}) = 0$ using a guess value $\vec{x}_0$.

Expanding $\vec{F}(\vec{x})$ in a Taylor series to linear order, we have,

$$\vec{F}(\vec{x} + \delta\vec{x}) = \vec{F}(\vec{x}) + J(\vec{x})\delta\vec{x}, \quad (1)$$

where $J(\vec{x})$ is the Jacobian matrix, i.e., $J_{ij} = \frac{\partial F_i}{\partial x_j}$. If $\vec{x} + \delta\vec{x}$ is a root, then the left hand side of the above equation must vanish. Then, we have,

$$J\delta\vec{x} = -\vec{F} \Rightarrow \delta\vec{x} = -J^{-1}\vec{F}. \quad (2)$$

Thus, the algorithm for finding the root is to change $\vec{x}$ following this rule until convergence is achieved.

$$\delta\vec{x}_{n+1} = \vec{x}_n - J^{-1}\vec{F}. \quad (3)$$

For a two-dimensional problem, J is a 2×2 matrix. Thus, if,

$$J(\vec{x}) = \begin{pmatrix} a(\vec{x}) & b(\vec{x}) \\ c(\vec{x}) & d(\vec{x}) \end{pmatrix}, \text{ then, } J^{-1}(\vec{x}) = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}. \quad (4)$$

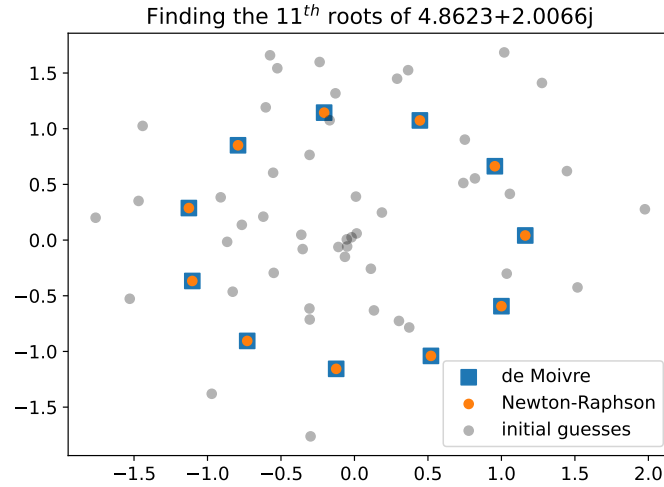


Figure 1: Sample calculation of roots with two-dimesional Newton-Raphson method and comparison with exact answers.

2 Improper Integrals

Consider the integrals, $I_c(R) = \oint_C f(z)dz$ and $I_r(R) = \int_{-R}^R f(x)dx$, C being a closed contour going from $-R$ to R along the real axis and coming back along a semi-circular contour of radius R centered at the origin and on the upper half of the complex plane. In many situations, the contribution to the semicircular part, $I_{sc}(R)$, vanishes as $R \rightarrow \infty$ and the value of the integral over the combined (closed) contour becomes equal to the

contribution from the real axis, i.e., $I \equiv \int_{-\infty}^{\infty} f(x)dx = \lim_{R \rightarrow \infty} I_r(R)$. In Fig. 2, for the function $f(x) = \frac{1}{1+x^2}$, the variation of the real and imaginary parts of the above integrals are plotted as a function of R .

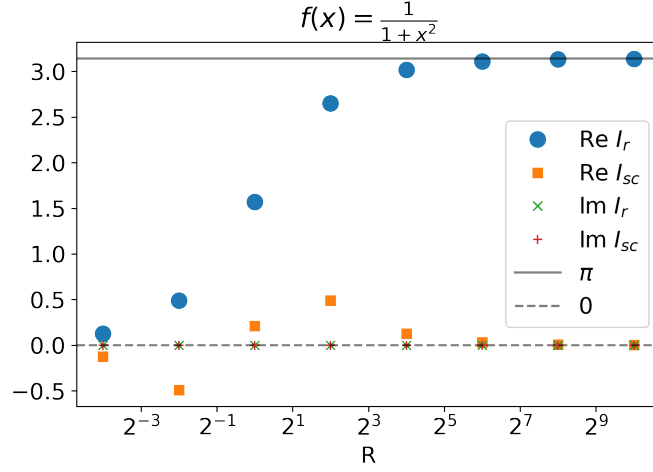


Figure 2: Variation of the real and imaginary parts of I_r and I_{sc} with R .

3 Gram-Schmidt Orthogonalization

Goal: Given a set of vectors, $\{\vec{x}_i, i = 0, 1, 2, \dots, n-1\}$, we need to find a set of orthogonal vectors, $\{\vec{x}_{io}, i = 0, 1, 2, \dots, n-1\}$, by forming suitable linear combinations of the vectors in the original set.

Algorithm

1. $\vec{x}_{0o} = \vec{x}_0$
2. For $i \geq 1$, $\vec{x}_{io} = \vec{x}_i - \sum_{j=0}^{i-1} \vec{x}_i \cdot \hat{x}_{jo} \cdot \hat{x}_{jo}$, i.e., we remove the components along the previously found vectors in the orthogonal set.

4 Gauss Elimination and Gauss-Jordan methods

Goal: Given an $n \times n$ matrix, $\mathbf{A}$, and an n -component vector $\mathbf{b}$, calculate $\det \mathbf{A}$ and solve the system $\mathbf{Ax} = \mathbf{b}$.

4.1 Gauss Elimination

Method: use row operations to make an upper triangular matrix and then back-substitute (just like we did in school with pen and paper).

Algorithm

1. Horizontally stack $\mathbf{A}$, and $\mathbf{b}$ and form the $n \times (n+1)$ matrix $\mathcal{A}$.
2. *Conditioning*
 - (a) For each row of $\mathbf{A}$, find the elements which are largest in absolute value. This forms the vector $\mathbf{a}$.
 - (b) Divide each row of $\mathcal{A}$ by the corresponding element of $\mathbf{a}$.
3. Let $s = 1$.
4. For i in $0, 1, 2, \dots, n-2$,
 - (a) *Pivot*: Find $j \geq i$, such that for the i^{th} column, excluding the elements above the diagonal, the j^{th} row has the largest element in absolute value.
 - (b) If the element which is largest in absolute value smaller than some threshold, e.g., 10^{-10} , then the matrix is singular. Stop with some error message.
 - (c) If $j > i$, swap the i^{th} and j^{th} rows and change the sign of s .

- (d) For j in $i + 1, i + 2, \dots, n - 1$, subtract from the j^{th} row, $\left(\frac{\mathcal{A}_{ij}}{\mathcal{A}_{ii}}\right)$ times the i^{th} row.
5. The value of the determinant is s times the product of the diagonal elements of $\mathcal{A}$ times the product of the elements of the vector $\mathbf{a}$.
6. *Back-substitution:*
- (a) $x_{n-1} = \frac{\mathcal{A}_{n-1,n}}{\mathcal{A}_{n-1,n-1}}$
- (b) For i in $n - 2, n - 1, \dots, 0$, $x_i = \frac{\mathcal{A}_{i,n} - \sum_{j=i+1}^{n-1} \mathcal{A}_{i,j}x_j}{\mathcal{A}_{i,i}}$

4.2 Gauss-Jordan method

Method: use row operations to make an identity matrix.

Algorithm

1. Horizontally stack $\mathbf{A}$, $\mathbf{b}$ and an $n \times n$ identity matrix to form the $n \times (2n + 1)$ matrix $\mathcal{A}$.
2. *Conditioning* Same as Gauss elimination.
3. Let $s = 1$.
4. For i in $0, 1, 2, \dots, n - 2$,
 - (a) *Pivot:* Find $j \geq i$, such that for the i^{th} column, excluding the elements above the diagonal, the j^{th} row has the largest element in absolute value.
 - (b) If the element which is largest in absolute value smaller than some threshold, e.g., 10^{-10} , then the matrix is singular. Stop with some error message.
 - (c) If $j > i$, swap the i^{th} and j^{th} rows and change the sign of s .
 - (d) Multiply each element of $\mathbf{a}$ with the corresponding diagonal element of $\mathcal{A}$.
 - (e) Divide the i^{th} row by $\mathcal{A}_{ii}$.
 - (f) For j in $0, 1, 2, \dots, i - 1, i + 1, i + 2, \dots, n - 1$, subtract from the j^{th} row, $\mathcal{A}_{ij}$ times the i^{th} row.
5. The value of the determinant is s times the product of the elements of the vector $\mathbf{a}$.
6. The last column of $\mathcal{A}$ gives the value of $\mathbf{x}$.
7. The previous n columns contain $\mathbf{A}^{-1}$.